

# Data-efficient cereal crop segmentation via mixture-of-experts fusion of vision foundation models

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**Abstract.** We present a data-efficient pipeline for semantic segmentation of cereal crops from Sentinel-2 imagery in Algeria. Leveraging two pretrained Vision Foundation Models (Prithvi-EO-2.0 and Satlas), we fine-tuned both under a few-shot regime (124 spatially disjoint image-mask pairs collected on site) and combined them through a lightweight **Mixture-of-Experts** (MoE) gating network. On a held-out validation split, we report mean Cereal-class Intersection-over-Union (IoU) =  $95.6 \pm 0.9\%$  and Overall Accuracy =  $97.8 \pm 0.6\%$  (mean  $\pm$  SD over 3 independent runs). Detailed comparative analysis shows that the MoE improves IoU by +1.5–2.0 % relative to the best single expert. We release code, model checkpoints, and a dataset sample to enable reproducibility. This work demonstrates that foundation-model fusion via MoE yields robust cereal mapping under severe label scarcity, with potential for transfer to other crops and regions.

**Keywords:** foundation models, few-shot learning, mixture of experts, crop mapping, Sentinel-2, semantic segmentation.

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## 1 Introduction

Agriculture represents a vital component of Algeria’s economy, contributing approximately 12% to the national GDP and employing nearly 25% of the workforce. However, the sector faces persistent challenges such as water scarcity, pest damage, and inefficient resource management, which collectively account for an estimated 20-30% loss of potential crop yields annually.<sup>1,2</sup> These constraints threaten national food security and underscore the need for innovative, technology-driven solutions.<sup>3</sup> Integrating remote sensing (RS) with artificial intelligence (AI) provides a transformative pathway for precision agriculture,<sup>4</sup> enabling data-driven monitoring of crop health and environmental conditions in real time.<sup>5,6</sup> Nevertheless, the effectiveness of such AI-based

monitoring remains limited by the scarcity of locally labeled datasets suitable for training robust deep learning models.

A fundamental impediment to deploying advanced AI in Algerian agriculture is the critical scarcity of labeled national datasets.<sup>7-9</sup> Given the limited availability of task-specific annotated data, developing deep learning models from scratch becomes impractical.<sup>10,11</sup> This data gap is further compounded by the high computational cost of training, with some models requiring over 100 GPU hours. These constraints strongly advocate for the use of data-efficient techniques such as transfer learning, which can leverage knowledge from large, pre-trained Vision Foundation Models (VFMs).<sup>12,13</sup>

Recent studies have further expanded the scope of agricultural monitoring through advanced deep learning frameworks and multi-sensor integration. For instance, the efficacy of ensemble learning for temporal mapping was demonstrated by utilizing a stacked machine learning model within Google Earth Engine to map cropping intensity patterns from combined Sentinel-2 and Landsat-8/9 time series.<sup>14</sup> In parallel, the importance of spectral preprocessing has been emphasized, showing that specific spectral enhancements on Sentinel-2 data significantly optimize deep learning-based segmentation performance for subtle crop and soil marks.<sup>15</sup> More recently, the applicability of foundation models has been validated through a fully automated framework using the Segment Anything Model (SAM) for rice seedling segmentation and health assessment, highlighting the potential of prompt-able architectures for precision agriculture.<sup>16</sup> These works collectively underscore the shift towards data-efficient, multi-modal, and foundation-model-driven approaches in modern remote sensing.

To address the aforementioned data scarcity, we collected a local dataset of 124 Sentinel-2 image-mask pairs representing cereal and non-cereal parcels in the Sétif region (Northeastern Algeria), using the Mergin Maps platform (Lutra Consulting, UK) for in-field polygon delineation between March and May 2024 (see Sec. 3.2). Each polygon was labeled as either 1 (cereal) or 0 (non-cereal), producing a binary segmentation dataset suitable for few-shot learning. This study directly explores the underutilized potential of fine-tuning VFMs through Few-Shot Learning (FSL)—a training approach that adapts models using only a small number of labeled examples—for robust semantic segmentation of crop types under severe data scarcity.<sup>17,18</sup> By developing accessible and computationally efficient monitoring solutions, our work aims to mitigate

inefficiencies in water and fertilizer management and support sustainable agricultural planning in resource-constrained environments.

The main contributions of this study are summarized as follows:

- We collected and curated a local Sentinel-2 dataset covering cereal crop parcels in the Sétif region of Algeria, providing 124 spatially disjoint image–mask pairs annotated for cereal versus non-cereal areas under severe data scarcity conditions.
- We fine-tuned two Vision Foundation Models, Satlas<sup>19</sup> and Prithvi-EO-2.0<sup>20</sup> and, using a few-shot learning strategy designed for limited samples while maintaining generalization across **unseen** parcels.
- We developed a lightweight MoE gating network that adaptively fuses predictions from both foundation models. This fusion strategy yielded superior segmentation accuracy (Cereal-Class IoU =  $95.6 \pm 0.9$  %), demonstrating improved robustness and stability compared to single-expert baselines. All code, trained weights, and a sample of the dataset are released to ensure reproducibility and future research reuse.

## 2 Related Work

### 2.1 Few-shot learning in remote sensing

Few-shot learning (FSL) and generalized few-shot semantic segmentation (GFSS) have received increasing attention in the RS community due to the persistent scarcity of dense annotations. Several recent benchmarks and methodological advances propose task-augmentation, prototype- and adapter-based strategies that leverage foundation models or learnable prompts to bridge domain gaps with very few labels.<sup>21,22</sup> Domain-specific techniques in RS—including label propagation, feature disentanglement, and multi-scale prototype aggregation—have delivered strong results on aerial and spaceborne few-shot segmentation benchmarks and motivate the use of few-shot fine-tuning for cereal mapping under severe label scarcity.<sup>17</sup>

### 2.2 Foundation models for Earth observation

Recent years have seen the rapid emergence of Vision Foundation Models (VFMs) tailored to Earth observation (EO), pretrained on very large collections of satellite and aerial imagery and then adapted to downstream tasks via transfer learning. Large-scale pretraining efforts such as SatlasPretrain have shown that pretraining on diverse, geospatially distributed imagery

substantially improves downstream performance on segmentation and mapping tasks.<sup>19</sup> New multitemporal foundation models (e.g., Prithvi-EO-2.0) further integrate temporal and location embeddings, making them particularly suitable for crop-mapping tasks.<sup>20</sup> Satellite-specific contrastive and vision-language approaches (e.g., SatCLIP) provide location-aware and open-vocabulary capabilities that enable stronger few-shot transfer in EO domains.<sup>23</sup> Transformer-based backbones and hybrid vision-time architectures (GeoViT and Geoformer) have also become popular choices for encoding the spatial and temporal structure of RS imagery.<sup>24,25</sup>

### *2.3 Ensembles, MoE, and diffusion-based approaches*

Ensemble learning and MoE frameworks have been widely explored to improve robustness and predictive accuracy in semantic segmentation tasks. Classical ensemble approaches in remote sensing typically rely on averaging or voting schemes across multiple models, such as U-Net variants with different backbones or training initializations. While these methods have demonstrated performance gains, they remain fundamentally static, as they assign equal or predefined importance to all models regardless of input characteristics.<sup>26,27</sup> This limitation reduces their effectiveness in heterogeneous agricultural landscapes where spectral and spatial variability can be significant.

MoE architectures address this limitation by introducing a learnable gating mechanism that dynamically weights the contribution of each expert model. Early work on MoE focused on large-scale sparsely gated networks for computational efficiency,<sup>28</sup> while more recent studies have adapted MoE concepts to dense prediction tasks, including semantic segmentation.<sup>27</sup> However, in the context of remote sensing, most MoE-related approaches remain limited in scope. Many existing implementations rely on relatively homogeneous experts (e.g., multiple CNNs with similar inductive biases) or require large-scale annotated datasets to train both experts and gating networks effectively. This makes them less suitable for scenarios characterized by severe data scarcity.

Furthermore, prior MoE-based segmentation approaches rarely explore the fusion of heterogeneous VFMs. Most existing works focus either on traditional deep learning architectures or on scaling single foundation models, rather than combining complementary pretrained models with distinct representational properties. As a result, the potential of MoE to exploit architectural diversity such as combining global attention mechanisms (ViT-based models) with hierarchical

local feature extractors (Swin-based models) remains underexplored in remote sensing applications.

In parallel, diffusion-based models have recently emerged as powerful tools in remote sensing, particularly for image generation, restoration, and spatiotemporal fusion.<sup>29–31</sup> While these approaches show promise for data augmentation and reconstruction, their application to semantic segmentation especially under few-shot conditions remains limited. Moreover, diffusion models typically introduce significant computational overhead, which may hinder their deployment in resource-constrained operational settings.

In summary, despite the progress in ensemble learning, MoE architectures, and generative modeling, several key gaps remain: (i) the lack of adaptive fusion strategies tailored to heterogeneous VFMs, (ii) limited applicability of existing methods under extreme label scarcity, and (iii) insufficient exploration of lightweight, computationally efficient MoE designs for operational remote sensing tasks. This study addresses these gaps by proposing a data-efficient MoE framework that dynamically combines two complementary foundation models Prithvi-EO-2.0 and Satlas under a strict few-shot regime.

A structured summary of recent representative works (2023–2025) is provided in Table 1. The table synthesizes key trends in foundation models, few-shot learning, and ensemble strategies, highlighting differences in data requirements, model design, and application scope, and providing context for the proposed MoE-based fusion approach.

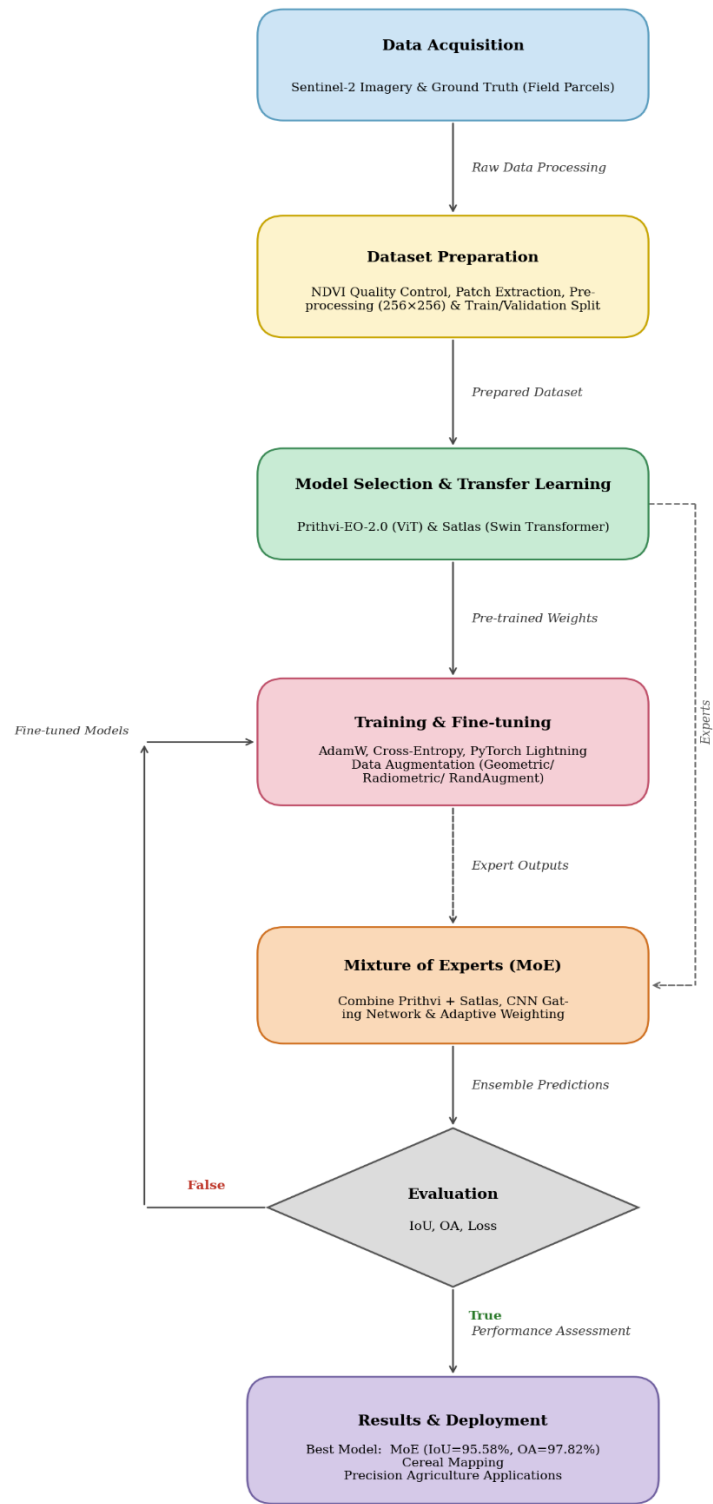
**Table 1** Recent works (2023–2025) related to foundation models, few-shot learning, ensemble/MoE and diffusion methods in remote sensing.

| Paper                        | Dataset(s)                    | Pretraining / Modality     | Task / Metric         | Key novelty                                                      |
|------------------------------|-------------------------------|----------------------------|-----------------------|------------------------------------------------------------------|
| Prithvi-EO-2.0 <sup>20</sup> | Harmonized Landsat + Sentinel | Multi-temporal pretraining | Segmentation / IoU    | Temporal + location embeddings; multi-resolution generalization. |
| GFSS <sup>21</sup>           | OpenEarthMap                  | —                          | Few-shot segmentation | Benchmark for generalized few-shot segmentation.                 |

|                               |                         |                            |                         |                                                      |
|-------------------------------|-------------------------|----------------------------|-------------------------|------------------------------------------------------|
| FM-Assisted FSL <sup>22</sup> | OpenEarthMap            | FM-assisted                | GFSS (mIoU)             | Foundation-model-assisted few-shot segmentation.     |
| GeoViT <sup>24</sup>          | Sentinel / multi-sensor | Transformer back- bone     | Segmentation / Accuracy | Vision transformer architecture for RS.              |
| Geoformer <sup>25</sup>       | Sentinel                | Transformer back- bone     | Prediction / MAE        | Vision and sequence transformer for RS.              |
| U-Net Ensemble <sup>26</sup>  | Various RS datasets     | Ensembles                  | Segmentation            | Ensemble gains via backbone diversity.               |
| RSDiff <sup>29</sup>          | Various RS imagery      | Diffusion-based generation | Image synthesis / SR    | Two-stage diffusion for high-res RS image synthesis. |
| CRS-Diff <sup>30</sup>        | RS benchmarks           | Diffusion tailored to RS   | Conditional generation  | Controllable diffusion for RS image generation.      |
| STFDiff <sup>31</sup>         | Spatiotemporal RS       | Diffusion spatiotemporal   | Fusion / reconstruction | Diffusion-based spatiotemporal fusion for RS.        |

### 3 Materials and Methods

Figure 1 provides an overview of the proposed methodology. The workflow begins with Sentinel-2 imagery acquisition during the field campaign conducted between March and May 2024, corresponding to the late tillering to early heading stages of cereal crops in the Sétif region. Field data were collected simultaneously using the Mergin Maps mobile app to delineate cereal parcels. Sentinel-2 Level-2A images from April and May 2024 were selected for their minimal cloud coverage and optimal phenological representation. The workflow proceeds with NDVI-based quality control and preprocessing. The processed dataset is then used to fine-tune two vision foundation models (Prithvi and Satlas), and finally, a MoE ensemble combines both models to generate accurate cereal mapping results. A detailed summary of the datasets utilized in this study is provided in Table 2. The detailed steps are described in the following subsections.



**Fig. 1** General workflow of the proposed methodology.

### 3.1 Study Area and Data Source

The primary study area is centered around Sétif (Fig. 2), a key agricultural zone in the Hautes Plaines region of northeastern Algeria, recognized for its significant cereal production. The study area spans latitudes 36.14°N–36.29°N and longitudes 5.27°E–5.48°E, covering part of the Sétif high plain at an average elevation of approximately 1,080 m above sea level. The region is characterized by a semi-arid Mediterranean climate (Köppen classification: Csa) with annual precipitation between 350–450 mm. The local agricultural calendar includes cereal sowing from November to December and harvesting between late May and June. Sentinel-2 Level-2A images were selected from April to May 2024, coinciding with the late tillering to early heading stages of cereals, when canopy contrast is optimal for spectral discrimination. Field surveys were conducted between March and May 2024 using the Mergin Maps mobile application to collect 124 annotated polygon parcels distinguishing cereal and non-cereal fields. Key metadata regarding the study area characteristics and data acquisition periods is summarized in Table 2.

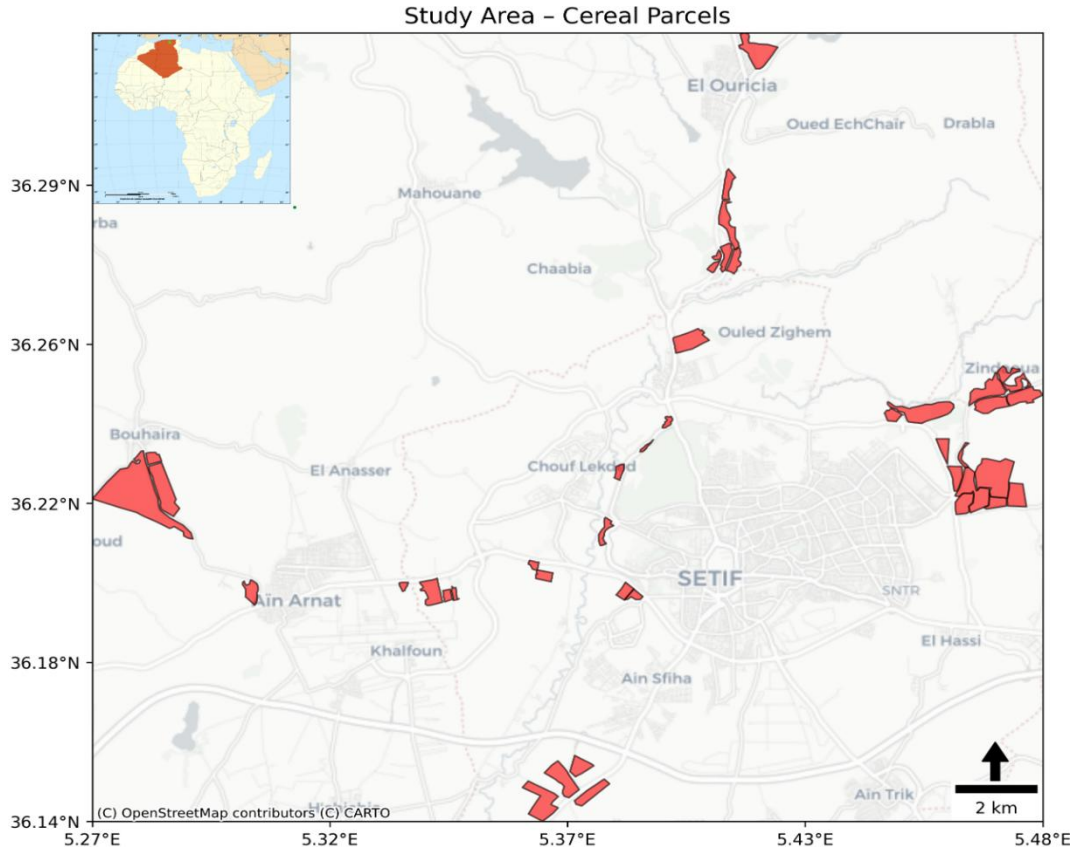
The remote sensing data were sourced from the Sentinel-2 mission,<sup>32</sup> utilizing Level-2A atmospherically corrected surface reflectance products accessed via Google Earth Engine (GEE). Six spectral bands were selected for their sensitivity to vegetation: Blue (B2), Green (B3), Red (B4), Near-Infrared (NIR, B8), and Short-Wave Infrared (SWIR1–B11, SWIR2–B12). These bands provide complementary information for crop monitoring, as they capture visible reflectance linked to plant chlorophyll, near-infrared sensitivity to vegetation structure and biomass, and short-wave infrared responses to leaf water content and soil moisture.<sup>33</sup> Red-edge bands (B5, B6, B7, B8A) were excluded to optimize computational efficiency, as prior studies showing that these bands provide only marginal improvements for crop-mapping tasks when NIR and SWIR channels are already included.<sup>34–36</sup> All Sentinel-2 scenes were atmospherically corrected and resampled to 10 m spatial resolution using bilinear interpolation.

**Table 2** Study area metadata.

| Parameter            | Value                          |
|----------------------|--------------------------------|
| Geographic extent    | 36.14°N–36.29°N, 5.27°E–5.48°E |
| Elevation            | ≈1080 m above sea level        |
| Climate              | Semi-arid Mediterranean (Csa)  |
| Mean annual rainfall | 350–450 mm                     |

|                  |                                             |
|------------------|---------------------------------------------|
| Main crop season | Nov–Jun (sowing: Nov–Dec, harvest: May–Jun) |
| Imagery period   | Apr–May 2024                                |
| Field campaign   | Mar–May 2024 (42 labeled parcels)           |

Figure 2 displays the mapped cereal parcels over the Sétif region, with geographic coordinates and scale reference.



**Fig. 2** The primary study area in Sétif, Algeria. The ground truth cereal parcel polygons collected for this research are highlighted in red.

### 3.2 Dataset Acquisition and Preprocessing

The datasets utilized in this study, comprising satellite imagery and field-collected ground truth records, are summarized in Table 3.

**Table 3** Summary of remote sensing and ground truth datasets used in this study.

| Product Name        | Spatial Resolution | Temporal Resolution | Periods Used   | Link/DOI                                                                                      |
|---------------------|--------------------|---------------------|----------------|-----------------------------------------------------------------------------------------------|
| Sentinel-2 Level-2A | 10 m/ 20m          | ~5 days             | April–May 2025 | <a href="https://doi.org/10.5281/zenodo.17674368">https://doi.org/10.5281/zenodo.17674368</a> |

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| Ground<br>Truth | Vector<br>(Polygons) | N/A | Mar–<br>May<br>2025 |
|-----------------|----------------------|-----|---------------------|
|-----------------|----------------------|-----|---------------------|

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Ground truth data were meticulously collected via three field campaigns conducted between March and May 2024 in Sétif using the Mergin Maps mobile application.<sup>37</sup> A team of two operators digitized cereal parcel polygons directly in the field using GPS-enabled smartphones with an average horizontal accuracy of  $\pm 2$  m.

Vegetation health and spectral dynamics were evaluated using time-series analysis of Sentinel-2 Level-2A surface reflectance products. To ensure phenological and spectral consistency, three vegetation indices were computed: the Normalized Difference Vegetation Index (NDVI), Normalized Difference Red Edge (NDRE), and Enhanced Vegetation Index 2 (EVI2). These were calculated as follows:

$$NDVI = \frac{NIR - Red}{NIR + Red}, \quad (1)$$

$$NDRE = \frac{NIR - RedEdge}{NIR + RedEdge}, \quad (2)$$

$$EVI2 = \frac{NIR - Red}{NIR + 2.4 \times Red + 1}, \quad (3)$$

where *NIR* corresponds to Band 8, *Red* to Band 4, and *RedEdge* to Band 5. Polygons were retained only if  $NDVI > 0.45$  and  $EVI2 > 0.35$  during the mid-season growth phase (April–May 2024). These thresholds were empirically established based on the observed distribution of field samples to rigorously distinguish active, healthy cereal stands at the heading stage from fallow land or sparse vegetation typical of semi-arid background noise. This multi-index filtering ensured that all parcels exhibited consistent vegetation vigor typical of active cereal stands.

The initial dataset contained 42 polygons. These underwent a rigorous quality assurance (QA) process based on geometric consistency, field notes, and the multi-index analysis described above. Six polygons exhibiting boundary ambiguity or anomalous phenological signatures were removed, yielding a final set of 36 validated polygons. To create the final dataset for training, these 36 polygons were sampled across multiple dates during the peak growing season, resulting in 124 distinct, spatially disjoint image-mask pairs, as illustrated in Fig. 3. Each polygon contributed between 2–5 temporal samples representing different acquisition dates, resulting in a temporally rich yet spatially non-overlapping dataset.

Finally, images and masks were cropped and resized to  $256 \times 256$  pixels. Data augmentation—including random rotations, flips, and brightness jittering—was applied to improve model robustness. All pixel values were normalized per spectral band using the mean ( $\mu$ ) and standard deviation ( $\sigma$ ) computed over the training set, according to Eq. 4:

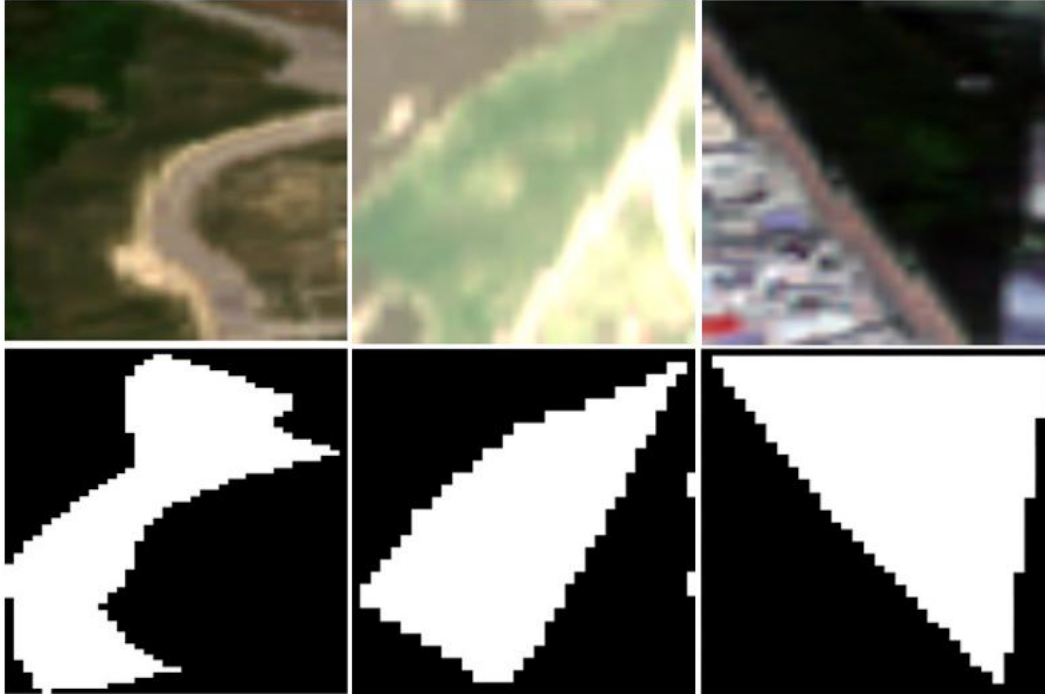
$$x'_c = \frac{x_c - \mu_c}{\sigma_c}, \quad (4)$$

where  $x_c$  denotes the pixel intensity in channel  $c$ . Table 4 lists the per-band statistics used for normalization. All bands were resampled to a uniform 10 m resolution.

**Table 4** Per-band reflectance statistics used for normalization.

| Band        | Mean ( $\mu$ ) | Std ( $\sigma$ ) |
|-------------|----------------|------------------|
| Blue (B2)   | 0.094          | 0.028            |
| Green (B3)  | 0.107          | 0.031            |
| Red (B4)    | 0.081          | 0.025            |
| NIR (B8)    | 0.312          | 0.072            |
| SWIR1 (B11) | 0.198          | 0.057            |
| SWIR2 (B12) | 0.142          | 0.048            |

The dataset was divided into 80% for training and 20% for validation using a polygon-level split. Crucially, all temporal acquisitions derived from a specific ground truth polygon were assigned exclusively to the same split (either training or validation). This protocol ensures zero spatial or temporal leakage between sets, maintaining the strict independence required for valid few-shot evaluation. To evaluate model stability, experiments were repeated across three independent runs, and performance metrics (IoU, OA) were reported as mean  $\pm$  standard deviation.



**Fig. 3** Examples of a training samples, showing the input Sentinel-2 image patch (top) and its corresponding binary ground truth mask (bottom), where white represents the 'cereal' class.

### 3.3 Foundation Models and Fine-Tuning

This research adopted a transfer learning approach to overcome the limitations of a modest dataset size.<sup>38</sup> Two VFMs were selected and fine-tuned for semantic segmentation:

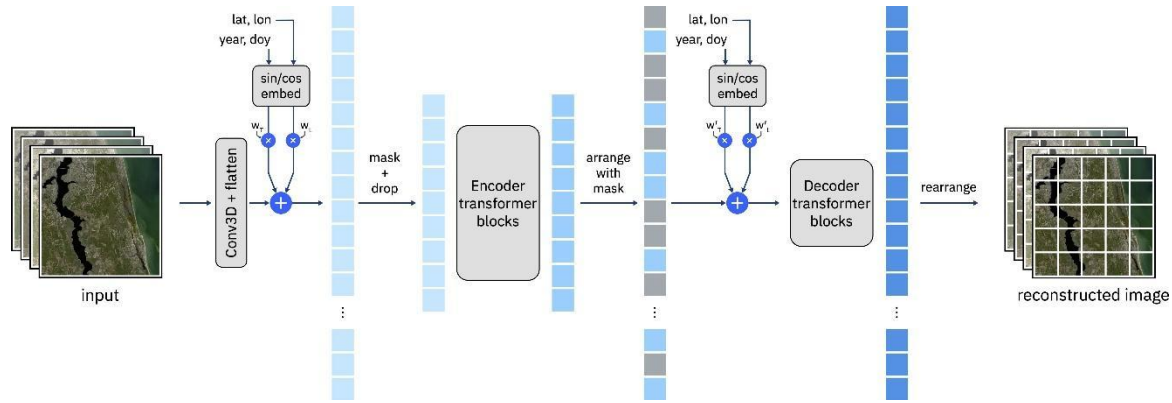
#### 3.3.1 Prithvi-EO-2.0

A ViT-based Earth Observation foundation model available in multiple sizes (notably 300 million and 600 million parameter variants). Prithvi-EO-2.0 is pretrained on multi-temporal harmonized Landsat+Sentinel-2 sequences (4.2M global time-series samples, ~30 m resolution) and incorporates explicit temporal and geolocation embeddings to model seasonal and geographic variability.<sup>19</sup> The model replaces standard 2D ViT patch embeddings with 3D spatiotemporal patches and uses 3D positional encodings generated from separate 1D sin/cos functions for time, height, and width. During pretraining, metadata such as acquisition date and location are encoded and added to token embeddings through learned weights, with random metadata-dropping applied to increase robustness. These architectural elements make Prithvi especially suitable for tasks where temporal dynamics (crop phenology) are informative.

Although Prithvi-EO-2.0 is inherently designed to leverage multi-temporal inputs, this study adopts a single-temporal ( $T = 1$ ) fine-tuning strategy. This design choice was primarily motivated by data constraints and experimental consistency. Specifically, the available dataset consists of a limited number of annotated parcels, and constructing temporally consistent multi-date sequences for each sample would significantly reduce the number of usable training instances, potentially compromising model generalization under a few-shot regime.

Furthermore, employing a single-temporal setup ensures a fair and controlled comparison with the Satlas model, which is also applied in a single-image configuration in this study. Introducing multi-temporal inputs for only one expert would introduce an additional confounding factor, making it difficult to attribute performance differences solely to architectural characteristics.

Nevertheless, incorporating multi-temporal sequences represents a promising direction for future work. Given the ability of Prithvi-EO-2.0 to explicitly model temporal dynamics, it is expected that leveraging multi-date inputs could further improve segmentation performance, particularly in capturing crop phenology and reducing ambiguity in challenging cases. This extension will be explored in future studies with larger and more temporally consistent datasets.



**Fig. 4** Architecture of Prithvi-EO-2.0.<sup>16</sup> The model uses ViT-based masked autoencoder pretraining with 3D patch embeddings to jointly encode spatial and temporal structure. Geolocation and temporal metadata are added through learned positional encodings.

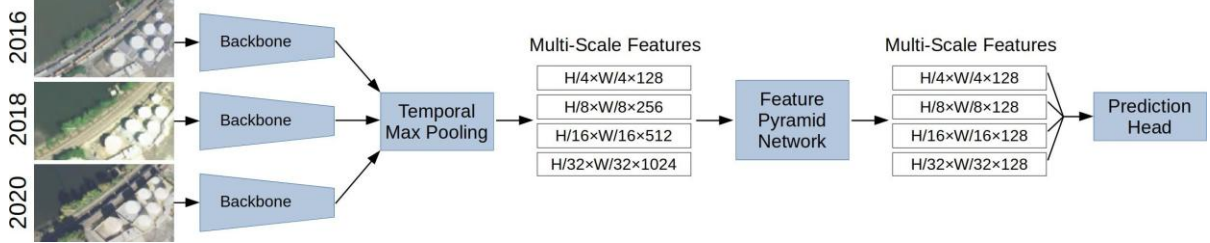
### 3.3.2 Satlas (SatlasPretrain models)

The models follow a modular Swin-Transformer architecture (Fig. 5) composed of a backbone, a feature pyramid network (FPN), and a prediction head.<sup>19</sup> The pretraining dataset combines Sentinel-2 and the US National Agriculture Imagery Program (NAIP) imagery paired with approximately 302M labels across 137 categories, generated from sources such as OpenStreetMap

and WorldCover together with curated annotations. For multi-image (temporal) inputs, the backbone processes each image independently and temporal max pooling aggregates representations across time. This supervised, large-scale pretraining strategy enables Satlas to learn strong spatial priors, feature hierarchies, and semantic regularities that transfer effectively to dense prediction tasks such as segmentation. To align with the pre-trained Satlas Sentinel-2 weights, our 6-band input was mapped to the required 9-channel format via a specific transformation  $T(x)$ :

- Channels 0-2 (RGB): B4, B3, B2.
- Channels 3-5: Duplicates of B4, B3, B2 (to satisfy architectural requirements).
- Channel 6: B8 (NIR).
- Channels 7-8: B11, B12 (SWIR). Input values were normalized and clipped to  $[0, 1]$ .

It is important to note that channels 3-5 were filled with duplicates of the RGB bands rather than available Red-Edge bands. This design choice was strictly dictated by the pre-training alignment of the Satlas backbone. The model weights for these input channels were optimized on RGB spectral distributions; replacing them with Red-Edge data—which exhibits significantly different radiometric characteristics—would introduce a severe spectral domain shift, invalidating the pre-trained feature extractors and degrading the transfer learning benefit.



**Fig. 5** Architecture of the SatlasPretrain model.<sup>19</sup> The backbone extracts spatial features from each input image, an FPN builds multi-scale feature representations, and the prediction head produces dense outputs. Multi-image inputs are aggregated using temporal max pooling.

Both foundation models provide complementary strengths: Prithvi captures temporal cues through explicit multi-date modeling, while Satlas provides strong supervised semantic representations due to its label-rich pretraining. In this study, both models were initialized from publicly available pretrained weights and fine-tuned on the Sentinel-2 training set described earlier.

Formally, the segmentation network  $f_\theta$  maps an input image patch  $x$  to per-pixel probabilities  $p$ , with binary predictions  $\hat{y}$  obtained as shown in Eq. 5:

$$p = f_\theta(x) \in [0,1]^{H \times W}, \quad \hat{y}_{ij} = 1\{p_{ij} \geq 0.5\}, \quad (5)$$

Training was conducted using the AdamW optimizer and the cross-entropy loss function (Eq. 6):

$$L_{CE} = -\frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W [y_{ij} \log \log p_{ij} + (1 - y_{ij}) \log \log (1 - p_{ij})], (6)$$

Where  $H \times W$  is the image patch size,  $y_{ij} \in \{0,1\}$  is the ground truth label at pixel  $(i,j)$ , and  $p_{ij} \in [0,1]$  is the model's predicted probability of that pixel belonging to the cereal class. The loss penalizes confident but wrong predictions (e.g.,  $y_{ij} = 0$  but  $p_{ij} \approx 1$ ).

While other studies have found success with CNN-based architectures like ResNet<sup>39</sup> and VGG<sup>40</sup>, this work focuses on Transformer-based VFMs. All training hyperparameters used for fine-tuning the Prithvi and Satlas models are summarized in Table 5.

**Table 5** Training hyperparameters used for fine-tuning the Prithvi and Satlas models.

| Hyperparameter     | Value                                  |
|--------------------|----------------------------------------|
| Learning rate      | 1e-4                                   |
| Batch size         | 8                                      |
| Number of epochs   | 50                                     |
| Optimizer          | AdamW                                  |
| Weight decay       | 1e-2                                   |
| Loss function      | Binary Cross-Entropy                   |
| Patch size         | 256 × 256 pixels                       |
| Scheduler          | None                                   |
| Data augmentation  | Rotation, flip, brightness jitter      |
| Normalization      | Per-band standardization (see Table 3) |
| Training framework | PyTorch Lightning                      |

### 3.3.3 Experimental Setup and Computational Efficiency

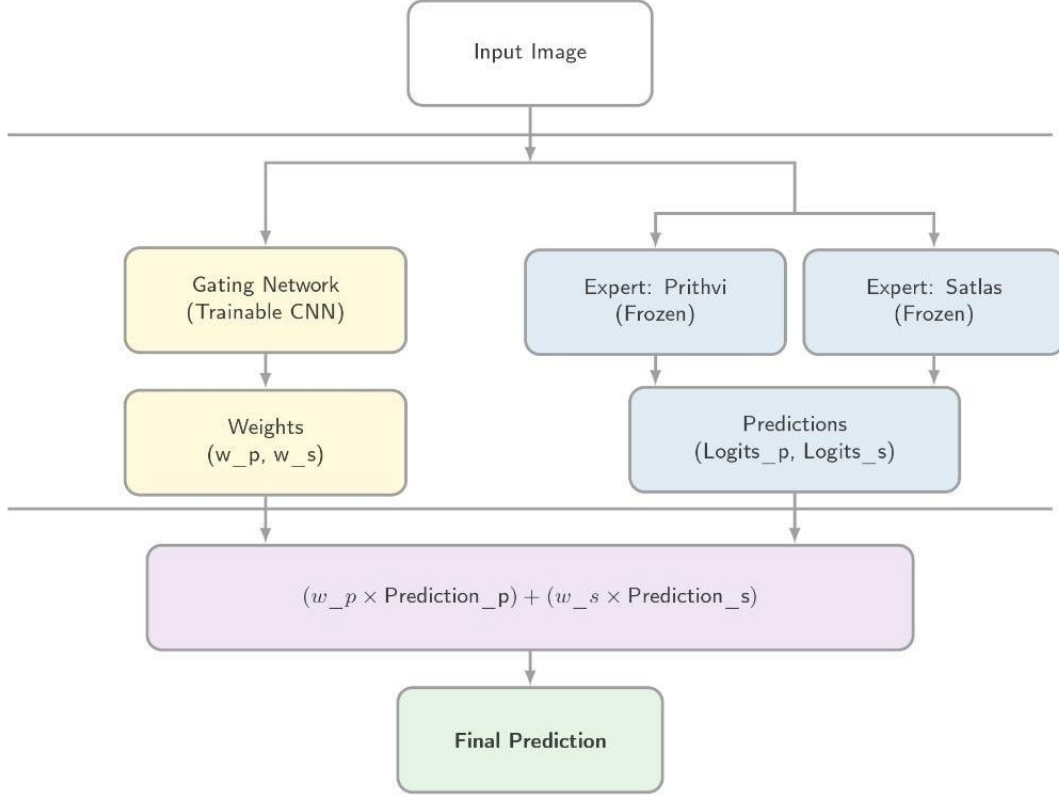
The fine-tuning experiments were conducted using NVIDIA GeForce RTX 1080 Ti GPUs. The training process for the MoE system, utilizing frozen experts, proved computationally efficient, requiring approximately 50 epochs to converge. Inference time for the complete MoE pipeline averages approximately 0.4 seconds per 256 × 256 patch, demonstrating the operational feasibility of the approach for regional-scale mapping.

### 3.4 Mixture of Experts

To potentially enhance performance beyond individual models, a MoE system was developed. MoE architectures are a powerful ensemble paradigm, originally introduced for efficiently scaling large neural networks by combining the strengths of multiple specialized sub-models (or “experts”).<sup>28</sup> The motivation for adopting an MoE approach in this study is primarily technical:

- **Architectural Complementarity:** The two chosen experts operate on distinct inductive biases. Prithvi-EO-2.0, being a standard Vision Transformer (ViT), utilizes global attention mechanisms that capture long-range semantic context across the patch. In contrast, Satlas employs a Swin Transformer backbone, which focuses on hierarchical, local window-based features. Fusing these two distinct representations allows the system to balance global scene understanding with fine-grained local texture recognition.
- **Robustness through ensembling:** Ensembles are known to reduce prediction variance and mitigate overfitting by combining multiple hypotheses, which is particularly valuable in limited-data regimes.
- **Dynamic weighting:** Unlike static averaging, the MoE framework employs a gating network that learns to adaptively assign weights to expert predictions depending on the input, enabling more flexible and context-aware decision-making.

The ensemble consists of the best-performing Prithvi and Satlas models, kept frozen as “experts”, and a lightweight, trainable CNN-based gating network, as represented in Fig. 6. The experts were kept frozen to preserve the robust feature representations learned during the individual fine-tuning stage and to avoid catastrophic forgetting given the limited dataset size.



**Fig. 6** Overall architecture of the proposed MoE system. The gating network receives the input Sentinel-2 patch and produces expert weights, which are used to compute a weighted combination of the Prithvi and Satlas expert predictions.

Formally, the final MoE prediction  $\hat{y}$  for an input image  $\mathbf{x}$  is obtained as a weighted combination of expert outputs (Eq. 7):

$$\hat{y} = \sum_{m=1}^M g_m(\mathbf{x}) f_m(\mathbf{x}), \quad (7)$$

Where  $f_m(\mathbf{x})$  denotes the output of the  $m$ -th expert,  $g_m(\mathbf{x})$  is the gating network weight for that expert, and  $M = 2$  in this study (Prithvi and Satlas). The gating weights satisfy  $\sum_{m=1}^M g_m(\mathbf{x}) = 1$ , ensuring a convex combination of expert predictions.

Unlike static fusion strategies such as uniform or fixed weighted averaging, the proposed MoE framework employs a dynamic gating mechanism that assigns input-dependent weights to each expert, enabling adaptive and context-aware fusion, we employ a compact two-layer CNN followed by global pooling and a fully connected layer. The complete architecture of this gating network is summarized in Table 6.

The gating network was trained end-to-end while keeping the expert models strictly frozen. We utilized the AdamW optimizer with a learning rate of  $1 \times 10^{-4}$  and weight decay of 0.01. Training proceeded for a maximum of 50 epochs with a batch size of 1 (due to memory constraints

of holding two experts) and Early Stopping monitored on validation loss with a patience of 10 epochs. To further improve robustness and generalization, the MoE system was trained using RandAugment<sup>41</sup>, a recently proposed data augmentation technique. Unlike fixed augmentation pipelines, RandAugment randomly selects augmentation operations (e.g., rotation, translation, color jitter, contrast adjustment) from a predefined pool and applies them with varying intensity. This stochasticity exposes the model to a wider diversity of transformations, reducing overfitting risks and enhancing the ability to generalize across unseen Sentinel-2 conditions and seasonal variations.

The gating network was intentionally designed as a lightweight convolutional architecture rather than a Transformer-based model. This design choice is motivated by three key considerations. First, given the extremely limited size of the training dataset (124 samples), a compact CNN provides stronger inductive biases for local spatial feature extraction and is less prone to overfitting compared to data-hungry Transformer architectures. Second, the role of the gating network in this framework is not to perform full semantic understanding, but rather to estimate coarse, global weighting coefficients between experts, for which high-capacity models are unnecessary. Third, maintaining a lightweight gating mechanism ensures computational efficiency, which is critical when combining multiple large foundation models during both training and inference. Preliminary experiments with deeper or more complex gating configurations did not yield consistent performance improvements, further supporting the suitability of a compact CNN design in this context.

The proposed gating network is considered lightweight in both architectural design and computational overhead. Specifically, it consists of a small number of convolutional layers followed by global pooling and a low-dimensional fully connected layer, resulting in a parameter count that is negligible compared to the underlying foundation models (Prithvi-EO-2.0 and Satlas), which contain tens to hundreds of millions of parameters. Consequently, the gating module contributes only a marginal increase in memory usage and inference time. From a computational perspective, the gating network operates on reduced spatial representations and produces only a small set of weighting coefficients, rather than dense pixel-level predictions. This significantly limits its computational cost relative to the expert models themselves. In contrast, alternative fusion strategies based on high-capacity architectures (e.g., Transformer-based gating or full feature-level fusion networks) would introduce substantially higher computational complexity and

increased risk of overfitting, particularly under few-shot conditions. Therefore, the proposed design achieves a favorable trade-off between adaptivity and efficiency, enabling dynamic expert selection while maintaining a low computational footprint suitable for practical remote sensing applications. In practice, the gating network represents less than 1% of the total parameter count of the combined model.

**Table 6** Architecture of the gating network used to compute expert weights in the Mixture-of-Experts system.

| Layer              | Configuration                                  | Output Shape                        |
|--------------------|------------------------------------------------|-------------------------------------|
| Input              | 6 input channels (Sentinel-2 bands)            | $B \times 6 \times H \times W$      |
| Conv2D + ReLU      | 16 filters, 3×3, stride 1, padding 1           | $B \times 16 \times H \times W$     |
| MaxPool2D          | 2×2 kernel, stride 2                           | $B \times 16 \times H/2 \times W/2$ |
| Conv2D + ReLU      | 32 filters, 3×3, stride 1, padding 1           | $B \times 32 \times H/2 \times W/2$ |
| MaxPool2D          | 2×2 kernel, stride 2                           | $B \times 32 \times H/4 \times W/4$ |
| Adaptive AvgPool2D | Output size: 1×1                               | $B \times 32 \times 1 \times 1$     |
| Flatten            | —                                              | $B \times 32$                       |
| Fully Connected    | Linear ( $32 \rightarrow N_{\text{experts}}$ ) | $B \times N_{\text{experts}}$       |
| Softmax            | Per-expert normalized weights                  | $B \times N_{\text{experts}}$       |

Algorithm 1 describes how the frozen Prithvi and Satlas experts are combined through the gating network during inference.

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**Algorithm 1.** Mixture-of-Experts (MoE) Inference and Gating

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**Input:** 6-channels Sentinel-2 patches  $\mathbf{x}$

**Input:** Frozen experts  $f_1$  (Prithvi),  $f_2$  (Satlas)

**Output:** Final fused logits  $\hat{\mathbf{y}}$

Compute gating weights:  $\mathbf{g} = \text{softmax}(\text{GatingCNN}(\mathbf{x}))$ ;

Run Prithvi expert:  $\mathbf{z}_1 = f_1(\mathbf{x})$ ;

Map  $\mathbf{x}$  to Satlas 9-channel format:  $\mathbf{x}_{\text{sat}} = T(\mathbf{x})$ ;

Run Satlas:  $\mathbf{z}_2 = f_2(\mathbf{x}_{\text{sat}})$ ;

Fuse outputs:

$$\hat{\mathbf{y}} = \mathbf{g}_1 \cdot \mathbf{z}_1 + \mathbf{g}_2 \cdot \mathbf{z}_2$$

**return**  $\hat{\mathbf{y}}$ ;

---

## 4 Results

Model performance was evaluated on the held-out validation set using standard semantic segmentation metrics: IoU, Overall Accuracy (OA), Precision, Recall, and the F1-score.<sup>42</sup> While

OA is reported for completeness, our analysis prioritizes the cereal-class IoU and F1-score, as these metrics provide a more robust evaluation of segmentation quality in the presence of class imbalance. All reported values are expressed as mean  $\pm$  standard deviation across three independent runs.

It is important to distinguish between two Intersection-over-Union (IoU) metrics reported in this study: (i) the macro-averaged IoU (mIoU), computed as the average IoU across all classes, and (ii) the class-specific IoU for the cereal class (IoU). While mIoU provides a global measure of segmentation performance, it can be influenced by class imbalance. In contrast, the Cereal-class IoU directly reflects the model’s ability to accurately delineate the target agricultural class of interest. For this reason, our analysis prioritizes the IoU as the primary evaluation metric, while mIoU is reported for completeness.

#### 4.1 Individual Model Performance and Ablation Study

To quantify the contribution of each component to the final system, we conducted an ablation study comparing the standalone performance of the fine-tuned foundation models against the fused Mixture-of-Experts architecture. The Prithvi model was evaluated under four configurations, varying the data augmentation strategy in each case (baseline, geometric augmentations, radiometric augmentations, and RandAugment). The RandAugment configuration achieved the best overall results, reaching  $81.42 \pm 0.48\%$  Overall Accuracy. However, the baseline Prithvi model achieved the highest IoU ( $70.49\%$ ), indicating that the choice of augmentation affects general accuracy and class-specific overlap differently.

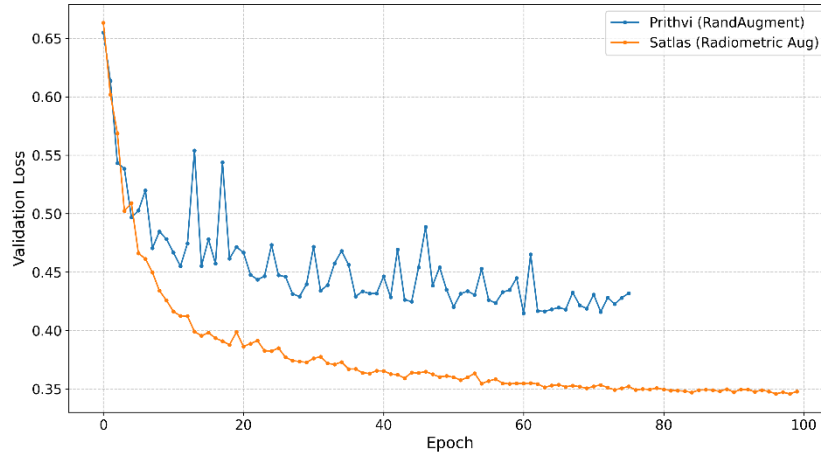
The Satlas model family demonstrated notably stronger performance. The best-performing variant was trained with subtle Radiometric Augmentation, achieving  $96.93 \pm 0.31\%$  Overall Accuracy and a IoU of  $94.12 \pm 0.33\%$ . This significantly surpassed the champion Prithvi configuration, as summarized in Table 7. Furthermore, Satlas achieved a high F1-score of  $96.92 \pm 0.28\%$ , reflecting highly consistent and reliable cereal discrimination.

**Table 7** Comparative Performance : Champion Prithvi (RandAugment) vs. Champion Satlas (Radiometric Augmentation).

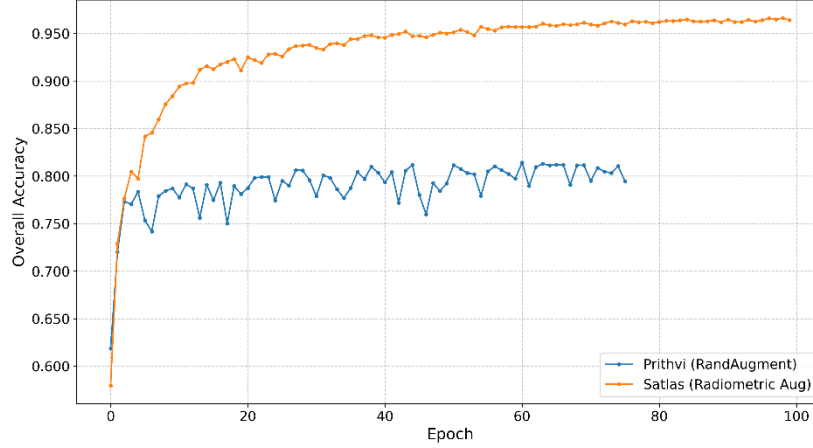
| Metric                   | Prithvi (RandAugment) | Satlas (Radiometric Aug.)          |
|--------------------------|-----------------------|------------------------------------|
| Overall Accuracy (%)     | $81.42 \pm 0.48$      | <b><math>96.93 \pm 0.31</math></b> |
| mIoU (Overall/Macro) (%) | $68.65 \pm 0.52$      | <b><math>94.04 \pm 0.36</math></b> |

|                        |                   |                                     |
|------------------------|-------------------|-------------------------------------|
| IoU (Cereal Class) (%) | $68.17 \pm 0.47$  | <b><math>94.12 \pm 0.33</math></b>  |
| Precision (%)          | $81.43 \pm 0.41$  | <b><math>96.88 \pm 0.33</math></b>  |
| Recall (%)             | $81.49 \pm 0.38$  | <b><math>96.97 \pm 0.29</math></b>  |
| F1 Score (%)           | $81.41 \pm 0.44$  | <b><math>96.92 \pm 0.28</math></b>  |
| Validation Loss        | $0.4146 \pm 0.35$ | <b><math>0.3427 \pm 0.29</math></b> |

To better illustrate the training dynamics of the two champion models, Fig. 7 and Fig. 8 show the validation loss and accuracy trajectories across training epochs. Both models exhibit smooth and stable convergence, indicating that the optimization process was well-behaved. The Satlas model reaches a lower validation loss and maintains consistently higher validation accuracy throughout training, which aligns with the superior final IoU and F1 scores reported in Table 8. These curves further confirm that the performance gap between Prithvi and Satlas is not due to instability or underfitting, but rather stems from differences in their underlying representational capacity and pretraining strategies.



**Fig. 7** Validation loss curves for the champion Prithvi (RandAugment) and Satlas (Radiometric Augmentation) models over training epochs. Both models exhibit stable convergence, with Satlas reaching consistently lower validation loss levels.



**Fig. 8** Validation accuracy curves for the champion Prithvi (RandAugment) and Satlas (Radiometric Augmentation) models. Satlas achieves higher validation accuracy throughout training, reflecting its stronger feature representation and more effective pretraining.

#### 4.2 Mixture of Experts (MoE) System Performance

While explicit baseline comparisons with static averaging schemes were not the primary focus of this study, the consistent improvement observed with the MoE approach over individual experts suggests that adaptive, input-dependent weighting provides a more effective fusion mechanism than fixed combination strategies commonly used in ensemble methods.

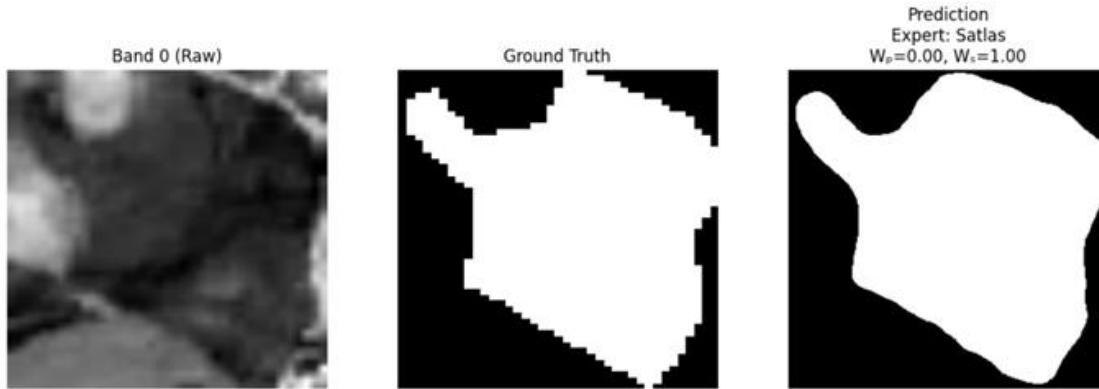
The MoE system successfully leveraged the complementary strengths of the two champion models (Prithvi-RandAugment and Satlas-Radiometric Augmentation). As detailed in Table 7, the MoE obtained the highest Overall Accuracy ( $97.82 \pm 0.26\%$ ) and IoU ( $95.58 \pm 0.29\%$ ). We observe that the standalone Satlas expert achieved marginally higher Precision (96.88%), Recall (96.97%), and F1-score (96.92%) compared to the MoE system. This indicates that while the supervised Satlas model is highly effective at pixel-level classification statistics, the MoE fusion provides a critical improvement in IoU (+1.46%). In agricultural mapping, where parcel boundaries are often irregular, IoU provides a more rigorous assessment of spatial adherence than the F1-score. Furthermore, the MoE demonstrated lower standard deviation across runs, indicating enhanced stability compared to the single-expert baselines.

The MoE further achieved a Precision of  $96.39 \pm 0.30\%$ , Recall of  $96.44 \pm 0.27\%$ , and an F1-score of  $96.41 \pm 0.28\%$ , indicating a highly consistent balance between detection reliability and completeness.

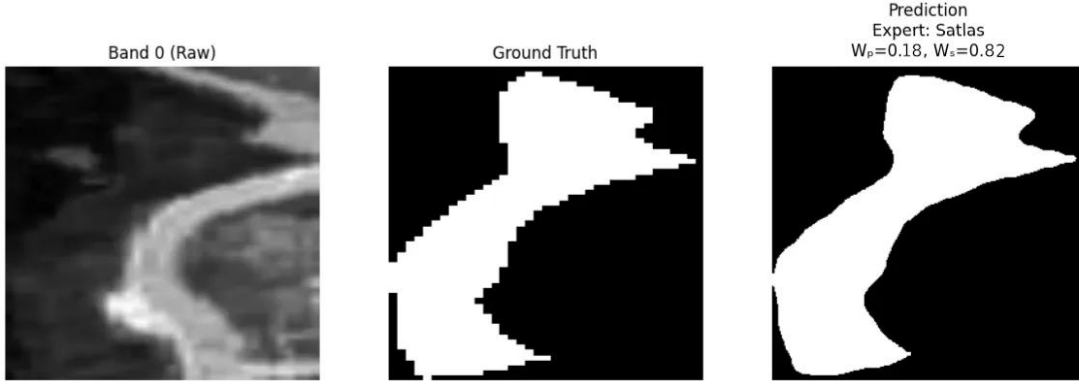
Analysis of the gating network revealed that it rapidly converged toward assigning higher weights to the Satlas expert—reflecting its stronger standalone performance—while still utilizing the Prithvi expert to refine difficult or ambiguous regions. This demonstrates that the MoE fusion mechanism effectively exploits cross-model complementarities rather than merely replicating the best single expert. Visual examples of these dynamic gating decisions are presented in Fig. 9, which illustrates qualitative segmentation results alongside the corresponding adaptive expert weights across different validation patches.

**Table 8** Comparative Performance : MoE System vs. Champion Models.

| Metric                 | Prithvi<br>(RandAugment) | Satlas (Radiometric<br>Aug.)       | MoE<br>System                       |
|------------------------|--------------------------|------------------------------------|-------------------------------------|
| Overall Accuracy (%)   | $81.42 \pm 0.48$         | $96.93 \pm 0.31$                   | <b><math>97.82 \pm 0.26</math></b>  |
| IoU (Cereal Class) (%) | $68.65 \pm 0.52$         | $94.04 \pm 0.36$                   | <b><math>95.58 \pm 0.29</math></b>  |
| Precision (%)          | $81.43 \pm 0.41$         | <b><math>96.88 \pm 0.33</math></b> | $96.39 \pm 0.30$                    |
| Recall (%)             | $81.49 \pm 0.38$         | <b><math>96.97 \pm 0.29</math></b> | $96.44 \pm 0.27$                    |
| F1 Score (%)           | $81.41 \pm 0.44$         | <b><math>96.92 \pm 0.28</math></b> | $96.41 \pm 0.28$                    |
| Validation Loss        | $0.4146 \pm 0.35$        | $0.3427 \pm 0.29$                  | <b><math>0.3341 \pm 0.31</math></b> |



(a) A patch where the gating network heavily favors the Satlas expert ( $W_p = 0.00, W_s = 1.00$ ).



(b) A patch where the gating network assigns mixed adaptive weights ( $W_p = 0.18, W_s = 0.82$ ).

**Fig. 9** Qualitative segmentation results from the final MoE model on validation patches demonstrating dynamic expert gating. From left to right: the input Sentinel-2 image, the ground truth mask, and the MoE predicted segmentation mask.

To contextualize these findings, we next compare our results with recent state-of-the-art studies.

## 5 Discussion

It is important to note that the experimental evaluation in this study is limited to a single geographic region and a single agricultural season; this aspect is discussed in more detail in the Limitations subsection.

The results obtained in this study demonstrate that foundation-model adaptation combined with a MoE fusion strategy can achieve highly accurate cereal crop mapping under severe data scarcity. Beyond reporting quantitative improvements, this section interprets the observed performance patterns in relation to existing deep learning approaches, analyzes sources of uncertainty affecting both data and models, and discusses the broader limitations and future directions of the proposed framework.

### 5.1 Comparison with Existing Deep Learning Architectures and Literature

The comparative performance of Prithvi-EO-2.0 and Satlas reflects their distinct pretraining strategies. Conventional deep neural networks trained from scratch often fail to generalize when

fewer than a few hundred labeled samples are available.<sup>9,10</sup> Therefore, we prioritized Vision Foundation Models (VFMs). Prithvi leverages masked autoencoder (MAE) self-supervision for general spectral–spatial representations,<sup>20</sup> whereas Satlas utilizes semantically rich supervised pretraining on over 302 million labels, optimizing it for dense-prediction tasks.<sup>19,43,44</sup> Beyond VFMs, robust architectures like EfficientNet (compound scaling),<sup>45</sup> SimPoolFormer (advanced token pooling)<sup>46</sup> and Tri-CNN (multi-branch spatial features)<sup>47</sup> achieve strong segmentation results. However, they typically require substantial annotated datasets or extensive tuning. In our extreme few-shot scenario (N=124), leveraging pretrained VFMs proved significantly more effective than training conventional architectures from scratch.

The proposed MoE framework further improved robustness by adaptively fusing the ViT-based Prithvi and Swin-based Satlas experts. The gating network assigned higher weights to Satlas while selectively activating Prithvi in ambiguous regions, acting as an implicit regularizer that outperforms simple averaging. An interesting nuance emerged in the metric trade-offs: while the MoE improved Overall Accuracy (+0.89%) and IoU (+1.46%) over the standalone Satlas model, it introduced a slight reduction in F1-score (from 96.91% to 96.41%). This occurred because the cross-entropy-trained gating network implicitly prioritized pixel-wise boundary alignment (directly benefiting IoU) over a strict precision-recall balance. Future MoE designs could incorporate multi-objective optimization to simultaneously maximize both metrics.

Compared to recent crop-mapping studies (Table 9), our MoE system achieves highly competitive results (95.58% IoU, 97.82% OA). While direct comparisons vary by dataset, recent foundation-model and classical deep learning approaches typically report mIoU values between 83–89% and 76–94%, respectively.<sup>48–52</sup> The superior IoU achieved here underscores the efficacy

of MoE-fused pretrained experts for fine-grained agricultural mapping under extreme label scarcity.

**Table 9** Comparison with selected recent crop-mapping studies.

| Study                                | Region / Data                   | Method                            | Metric                       | Reported Value                         |
|--------------------------------------|---------------------------------|-----------------------------------|------------------------------|----------------------------------------|
| Hosseini et al., 2024 <sup>14</sup>  | Iran (Sentinel-2 + Landsat-8/9) | Stacked Ensemble                  | OA; Kappa; F1                | 90.97–94.24%; 0.87–0.92; 71.87–98.27%  |
| Agapiou, 2025 <sup>15</sup>          | Italy (Sentinel-2)              | U-Net (mit_b5 + SCSE); YOLOv8-seg | IoU; F1-score                | 2–22% (IoU); 8–36% (F1)                |
| Rezvan et al., 2025 <sup>16</sup>    | Taiwan & China (UAV RGB)        | SAM + OCSVM                       | Dice Score; Silhouette Score | 72.6–94.7%; 0.31–0.44                  |
| Mohammedi et al., 2024 <sup>17</sup> | France / Ghana (S2 time series) | Few-shot learning (-TIM)          | Macro-F1                     | 75.9% (FR); 59.6% (GH)                 |
| Gao et al., 2023 <sup>48</sup>       | USA (Sentinel-2)                | A2SegNet                          | OA; mIoU gain                | 91.5–94.2% (OA); +6.6% (mIoU)          |
| Lu et al., 2023 <sup>49</sup>        | Sentinel-2                      | MSSNet                            | OA; mIoU                     | 90.77% (OA); 76.59 (mIoU)              |
| Yan et al., 2025 <sup>50</sup>       | PASTIS-R benchmark              | Spatiotemporal GFM                | mIoU; F1                     | 84–89% (mIoU) ; Up to 92% (F1)         |
| Li et al., 2025 <sup>51</sup>        | Multi-sensor (MODIS, L8/9, S2)  | Multi-source temporal FM          | mIoU                         | 83–88% (mIoU on downstream crop tasks) |
| Zhu et al., 2024 <sup>52</sup>       | SITS segmentation               | Weakly-sup. temp. net             | IoU-equiv.; F1               | 90–94% (IoU-equiv.); 92% (F1)          |

## 5.2 Sources of Uncertainty and Model Sensitivity

Despite the strong quantitative performance, several sources of uncertainty influence both the input data and the modeling process and must be considered when interpreting the results.

### 5.2.1 Input Data Uncertainty

Remote sensing-based crop mapping inherently involves uncertainties related to sensor characteristics and environmental variability. Sentinel-2 imagery, although providing valuable spectral information, has a spatial resolution of 10 m, which may introduce mixed-pixel effects

along parcel boundaries or in fragmented agricultural landscapes. Such spatial averaging can blur transitions between cereal and non-cereal classes, potentially affecting segmentation accuracy.

Atmospheric conditions represent another source of uncertainty. Although Level-2A products provide atmospheric correction, residual effects from aerosols, illumination variability, and seasonal atmospheric differences may still influence spectral reflectance. Additionally, cloud contamination and thin cirrus layers even when partially filtered can introduce subtle radiometric inconsistencies that propagate into model predictions.

Phenological variability also contributes to uncertainty. Crop appearance changes across growth stages, and although images were selected during optimal phenological windows (late tillering to early heading), inter-field differences in sowing dates, irrigation practices, or soil moisture conditions may alter spectral signatures. The NDVI and EVI2 filtering strategy reduces this variability but cannot completely eliminate intra-class heterogeneity.

Ground truth annotation introduces additional uncertainty. Field polygons were digitized using GPS-enabled smartphones with approximately  $\pm 2$  m positional accuracy, meaning that boundary pixels may contain labeling imprecision relative to satellite resolution. Such uncertainty is common in agricultural mapping and may partly explain residual prediction errors near field edges.

### *5.2.2 Model and Training Uncertainty*

Model performance in few-shot learning scenarios is inherently sensitive to training configuration choices. Hyperparameters such as learning rate, augmentation intensity, and optimizer settings influence convergence behavior and generalization. The observed differences among Prithvi augmentation strategies demonstrate that data augmentation plays a critical role in stabilizing training under limited data availability.

Stochastic optimization further introduces variability between runs. Reporting results as mean  $\pm$  standard deviation across three independent experiments partially addresses this issue, providing

a more reliable estimate of model stability. Nevertheless, small datasets inherently amplify sensitivity to initialization and sampling effects.

The MoE gating mechanism introduces another layer of uncertainty related to expert weighting. While the gating network consistently favored Satlas predictions, slight variations in gating weights may occur depending on input characteristics. However, this adaptive weighting mechanism also contributes to robustness by allowing dynamic correction of expert-specific errors.

### *5.3 Limitations and Future Research Directions*

The primary limitation of this study is its restricted geographic extent. All experiments were conducted on a dataset collected entirely within a single agroecological region (Sétif, Algeria). Consequently, the statistical significance of the performance improvement gained by utilizing the MoE approach requires further validation. Evaluating this architectural fusion on several geographically distinct datasets is necessary to conclusively prove robust statistical significance and generalizability under different climate regimes and soil management practices.

Future research will prioritize: (i) cross-dataset and cross-regional validation; (ii) evaluating the pipeline using alternative sensors (e.g., Landsat-9, PlanetScope) to assess sensor-agnostic performance; and (iii) investigating the MoE gating network's scalability when integrating additional domain-specific expert models.

Another limitation is the use of single-temporal inputs for Prithvi-EO-2.0, which does not fully exploit its temporal modeling capabilities. Future work will investigate multi-temporal extensions to better capture crop phenology and improve robustness across varying growth stages.

## **6 Conclusion**

This study successfully demonstrated the viability of data-efficient cereal crop mapping in label-scarce regions by employing a lightweight Mixture-of-Experts (MoE) gating network to adaptively fuse pre-trained Vision Foundation Models (VFMs). By leveraging only 124 carefully curated Sentinel-2 image-mask pairs from the Sétif province of Algeria, we established that:

- **Pretraining Efficacy:** VFMs pretrained on large-scale geospatial corpora (Satlas) transfer more effectively to few-shot semantic segmentation tasks than those relying on self-supervised reconstruction objectives (Prithvi-EO-2.0).
- **Superior Boundary Delineation:** The proposed MoE architecture effectively combined the complementary feature representations of Swin-Transformer (Satlas) and ViT-based (Prithvi) backbones, yielding superior operational performance. This is evidenced by a decisive Cereal-Class IoU of 95.58% and an Overall Accuracy of 97.82%, representing a substantial +1.54% IoU improvement over the champion single-expert baseline.
- **Operational Viability and Open Data:** Although the fusion mechanism prioritizes tighter spatial delineations, resulting in a slightly lower pixel-wise F1-score (96.41%) compared to the single Satlas expert (96.92%), this trade-off is operationally superior for precision agricultural applications. Confirming that adaptive foundation-model fusion is a highly robust solution for severe data-scarcity regimes, we highlight the release of our CerealMapSet-DZ dataset and validated implementation code to provide a foundation for future research into cross-regional validation and sensor-agnostic performance for global monitoring.

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## *Disclosures*

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper.

## Code, Data, and Materials Availability

The source code, models, and dataset generated and analyzed during the current study are openly available in a GitHub repository at the following URL: <https://github.com/Amir-001/Final-Year-Project>. and a Zenodo DOI: <https://doi.org/10.5281/zenodo.17674368>.

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## **Caption List**

**Fig. 1** General workflow of the proposed methodology.

**Fig. 2** The primary study area in Sétif, Algeria. The ground truth cereal parcel polygons collected for this research are highlighted in red.

**Fig. 3** Examples of a training samples, showing the input Sentinel-2 image patch (left) and its corresponding binary ground truth mask (right), where white represents the 'cereal' class.

**Fig. 4** Architecture of Prithvi-EO-2.0. The model uses ViT-based masked autoencoder pretraining with 3D patch embeddings to jointly encode spatial and temporal structure. Geolocation and temporal metadata are added through learned positional encodings.

**Fig. 5** Architecture of the SatlasPretrain model. The backbone extracts spatial features from each input image, an FPN builds multi-scale feature representations, and the prediction head produces dense outputs. Multi-image inputs are aggregated using temporal max pooling.

**Fig. 6** Overall architecture of the proposed Mixture-of-Experts (MoE) system. The gating network receives the input Sentinel-2 patch and produces expert weights, which are used to compute a weighted combination of the Prithvi and Satlas expert predictions.

**Fig. 7** Validation loss curves for the champion Prithvi (RandAugment) and Satlas (Radiometric Augmentation) models over training epochs. Both models exhibit stable convergence, with Satlas reaching consistently lower validation loss levels.

**Fig. 8** Validation accuracy curves for the champion Prithvi (RandAugment) and Satlas (Radiometric Augmentation) models. Satlas achieves higher validation accuracy throughout training, reflecting its stronger feature representation and more effective pretraining.

**Fig. 9** Qualitative segmentation result from the final MoE model on a sample validation patch. From left to right: the input image, the ground truth mask, and the model's predicted segmentation mask. **(a)** A patch where the gating network heavily favors the Satlas expert ( $W_p = 0.00, W_s = 1.00$ ). **(b)** A patch where the gating network assigns mixed adaptive weights ( $W_p = 0.18, W_s = 0.82$ ).

**Table 1** Recent works (2023–2025) related to foundation models, few-shot learning, and ensemble/MoE and diffusion methods in remote sensing.

**Table 2** Study area metadata.

**Table 3** Per-band reflectance statistics used for normalization.

**Table 4** Training hyperparameters used for fine-tuning the Prithvi and Satlas models.

802   **Table 5** Architecture of the gating network used to compute expert weights in the Mixture-of-  
803   Experts system.

804   **Table 6** Comparative Performance : Champion Prithvi (RandAugment) vs. Champion Satlas  
805   (Radiometric Augmentation).

806   **Table 7** Comparative Performance: MoE System vs. Champion Models.

807   **Table 8** Comparison with selected recent crop-mapping studies.